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Pediatric basic life support (BLS) for health care providers

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All topics are updated as new evidence becomes available and our peer review process is complete.

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INTRODUCTION

This topic review addresses general pediatric basic life support (BLS) principles for health care providers. Specific pediatric topics that include BLS management or opioid-associated emergencies are discussed separately:

- (See "Basic airway management in children" and "Airway foreign bodies in children", section on 'Foreign body removal'.)
- (See "Neonatal resuscitation in the delivery room".)
- (See "Opioid intoxication in children and adolescents".)

EPIDEMIOLOGY AND SURVIVAL

Early recognition and treatment of cardiac arrest improves survival for children and adults [1-3]. Effective pediatric BLS by trained health care providers or lay rescuers is the foundation of successful resuscitation [4].

Cardiac arrest — Cardiac arrest is a condition defined by the absence of pulses. The two types of pediatric cardiac arrest are hypoxic/asphyxial arrest and sudden cardiac arrest.

- **Hypoxic/asphyxial arrest** – Hypoxic/asphyxial arrest is the most common type of cardiac arrest in infants and children. It results from progressive tissue hypoxia and acidosis due to respiratory failure and/or shock [5]. Causes of respiratory failure and/or hypotensive shock leading to cardiac arrest in these age groups include septic shock, upper airway obstruction, pneumonia, sudden infant death syndrome, metabolic derangement (eg, hypoglycemia, inborn errors of metabolism, or electrolyte disturbance), neurological catastrophe (brain herniation or hemorrhage), poisoning (eg, opioid overdose), trauma, non-trauma exsanguination, and anaphylaxis [1,6-10].

  This finding is in contrast to adults, for whom the most common cause of cardiac arrest is ischemic cardiovascular disease. (See "Adult basic life support (BLS) for health care providers", section on 'Epidemiology and survival'.)

- **Sudden cardiac arrest** – Sudden cardiac arrest, most often caused by the development of ventricular fibrillation (VF) or pulseless ventricular tachycardia (pVT), is less common in infants and children than in

adults. Predisposing conditions or causes of ventricular arrhythmias in pediatric patients with sudden cardiac arrest include:

- Hypertrophic cardiomyopathy (see "Hypertrophic cardiomyopathy in children: Clinical manifestations and diagnosis")
- Myocarditis (see "Clinical manifestations and diagnosis of myocarditis in children")
- Anomalous coronary artery (from the pulmonary artery) (see "Congenital and pediatric coronary artery abnormalities", section on 'Variations of coronary artery origin from the pulmonary artery')
- Wolff-Parkinson-White syndrome (see "Wolff-Parkinson-White syndrome: Anatomy, epidemiology, clinical manifestations, and diagnosis")
- Long QT syndromes (see "Congenital long QT syndrome: Epidemiology and clinical manifestations" and "Acquired long QT syndrome: Clinical manifestations, diagnosis, and management")
- Drug intoxication (table 1)
- Commotio cordis (ie, sharp blow to chest) (see "Commotio cordis")

- **Respiratory arrest versus cardiac arrest** – Infants and children with a respiratory arrest (not breathing normally but presence of pulses), are much more likely to recover if promptly treated than those with a cardiac arrest [1,11-13]. As an example, in an observational study of 95 children with out-of-hospital cardiac arrest, 82 percent of children with respiratory arrest were alive at one year compared with 14 percent of patients with a cardiac arrest [1].

Individual determinants of survival following pediatric cardiac arrest vary according to the setting (out of hospital versus in hospital) and individual patient factors as discussed below.

Out-of-hospital cardiac arrest — The majority of pediatric out-of-hospital cardiac arrests (OHCA) are unwitnessed, are nontraumatic, occur at home, and occur in infants [9,10,14-18].

Survival to hospital discharge in pediatric OHCA is approximately 11 percent, with overall survival (and survival with good neurological function) varying by age [10]:

- Infants: 7 percent (5 percent)
- Children: 16 percent (13 percent)
- Adolescents: 19 percent (17 percent)

Survival from pediatric OHCA has not changed significantly over time or by region: the 7 to 10 percent overall survival from pediatric OHCA reported by a North American multicenter OHCA registry during 2007 to 2012 is consistent with data reported by other national registries [16,19-21].

Factors associated with increased survival after pediatric OHCA include [15,19,22]:

- Older age
- Witnessed arrest
- High-quality BLS including bystander cardiopulmonary resuscitation (CPR) and telephone dispatcher-assisted CPR (DA-CPR) [17,23-26]
- Automated external defibrillator (AED) use
- Initial shockable rhythm [27]

- Early presence of emergency medical services (EMS)
- Non-traumatic etiology [17]

Sudden cardiac arrest due to a shockable rhythm (VF or pVT) occurs in up to 18 percent of all pediatric OHCAs but is less commonly the presenting rhythm in children one to eight years of age (8 percent) and infants [10,28,29].

Early CPR after a witnessed cardiac arrest and receipt of public access defibrillation (PAD) are key interventions that improve neurologic outcome [1,22,28,30-33]. As an example, in a registry study of 188 children with OHCA during school sports (87 percent receiving bystander CPR), 66 percent of the 119 patients who received PAD had favorable neurologic outcome compared with 30 percent who did not [34].

In-hospital cardiac arrest — Survival to hospital discharge in pediatric in-hospital cardiopulmonary arrest (IHCA) is approximately 49 percent (overall) and 41 percent (pulseless arrest) [10]. Adjusted probability of survival (and favorable neurological outcome among survivors) was 41 percent (70 percent) for CPR duration of 1 to 15 minutes and 12 percent (60 percent) for CPR duration greater than 35 minutes [35].

Rates of survival from pediatric cardiac arrest are higher for IHCA [11,12,36-41] than OHCA [1,7-10,14,22,28,30-33]. A narrative literature review of this topic until January 2019 cited a threefold higher rate of survival among infants and children with IHCA compared with OHCA.

Increased survival after pediatric IHCA is associated with [15]:

- Younger age (lower mortality despite higher incidence in infants)
- Shorter duration of CPR
- Initial shockable rhythm
- Previous monitoring
- Occurrence on weekdays and during daytime

Longitudinal studies from the United States have identified a trend towards improved survival after pediatric IHCA [10-12,38]. According to the "Get With the Guidelines" (GWTG) resuscitation registry during 2000 to 2018, survival to discharge increased 29 to 41 percent for all IHCA, 19 to 41 percent for pulseless arrest, and 57 to 66 percent for non-pulseless arrest [10,12].

- During 2000 to 2009, the GWTG registry demonstrated an increase in survival to discharge from IHCA while the rate of significant neurologic disability remained stable [11]. Survival improved despite a high prevalence (in up to 85 percent patients) of an initial non-shockable rhythm (asystole and pulseless electrical activity [PEA]) and was similar regardless of the initial rhythm (shockable or non-shockable) [11].

- During 1997 to 2012, a retrospective cohort study of almost 30,000 discharges after IHCA demonstrated that the incidence of IHCA almost doubled to one IHCA per 1000 discharges while survival after IHCA increased from 49 to 60 percent [38].

During 2018, the GWTG registry reported the prevalence of initial rhythm for IHCA in 571 patients [10]:

- Shockable: 9 percent
- Non-shockable: 77 percent (asystole: 28.5 percent; PEA: 48.5 percent)
- Unknown: 14 percent

Taken together, these findings suggest that rapid recognition of cardiac arrest and prompt performance of CPR during in-hospital resuscitation are important contributors to improved survival in IHCA, and that, unlike patients with OHCA, survival from IHCA is less dependent upon the presenting rhythm.

Additional studies have evaluated the contribution of advanced techniques, such as endotracheal intubation and extracorporeal membrane oxygenation (ECMO) during resuscitation and post-resuscitation care, to survival after IHCA:

- **Endotracheal intubation** – In a retrospective observational study of over 2200 United States children younger than 18 years of age with IHCA reported to a centralized registry from 2000 to 2014, endotracheal intubation during cardiac arrest was associated with a significantly lower adjusted survival to hospital discharge compared with a propensity-matched cohort of patients who were not intubated (36 versus 41 percent, respectively) [39]. There was no significant difference in return of spontaneous circulation (ROSC) or favorable neurologic outcome between the groups. Although confounding cannot be fully excluded given the study design, this evidence suggests that the emphasis on early endotracheal intubation during pediatric IHCA warrants re-examination and further study [40].

- **ECMO during resuscitation and post-resuscitation care** – Use of ECMO during resuscitation and post-resuscitation care in settings with existing ECMO protocols, expertise, and equipment may be beneficial for selected patients who fail conventional CPR after IHCA, especially children with underlying cardiac disease such as cardiomyopathy, myocarditis, or congenital heart disease. Survival after ECMO during resuscitation and post-resuscitation care is discussed in greater detail separately. (See "Pediatric advanced life support (PALS)", section on 'Extracorporeal membrane oxygenation (ECMO) with CPR (ECPR)'.)

Bradycardia with poor perfusion — Bradycardia with poor perfusion (BPP) is a condition defined by heart rate <60/minute in a patient with cardiorespiratory compromise.

Evidence for hospital and pre-hospital treatment of BPP is as follows:

- **In-hospital bradycardia with poor perfusion** – In prospective, multicenter observational studies, the majority of children who receive CPR in the ICU setting have an initial rhythm of BPP [41-43]. Of these, the majority develop pulselessness [44].

  Compared to pulseless arrest, children with BPP in the ICU setting who receive CPR are younger and more likely to [43]:

  - Have an underlying respiratory etiology [43]
  - Survive to hospital discharge [11,12,42,43,45]
  - Survive with favorable neurologic outcome [43]

  CPR followed by bolus doses of epinephrine is recommended for children with BPP [44,46]. However, bolus epinephrine in these patients has not been associated with favorable outcomes. For example, in a secondary analysis of a prospective multicenter study of pediatric in-hospital CPR events in which 322 children with an initial rhythm of BPP received epinephrine within the first 2 minutes of starting CPR, bolus epinephrine was not associated with survival to hospital discharge or survival with favorable neurologic outcome [44]. ROSC was highest for patients who were never pulseless (85 percent) compared with BPP progressing to pulseless arrest (43 percent). A limitation of this study is that the early epinephrine group may have had a higher severity of illness.

- **Out-of-hospital bradycardia with poor perfusion** – Evidence suggests that bradycardia with poor perfusion in the out-of-hospital setting is undertreated. For example, in a retrospective study from the United States of 911 scene response encounters for 1209 children (median age two years) who met current PALS criteria of bradycardia with poor perfusion (first heart rate <60/minute with altered mental status or hypotension for age), only 12.1 percent received bag-mask ventilation (the most frequently performed airway intervention), 14.1 percent received CPR, 35 percent received epinephrine, 1.2 percent received atropine, and 24.7 percent received fluids [47]. Receipt of any PALS-recommended intervention was associated with altered mental status and age-adjusted hypotension, but not with first heart rate <60/minute). Survival outcomes were not reported. This study supports continued education and assessment of US prehospital providers regarding indications for PALS-recommended interventions for bradycardia with poor perfusion.

INTERNATIONAL RESUSCITATION GUIDELINES

Based upon extensive review of clinical and laboratory evidence, the American Heart Association (AHA) has published frequent updates for pediatric BLS [4,46,48-50], and the International Liaison Committee on Resuscitation (ILCOR) has published annual treatment recommendations [51-54].

BLS guidelines differ according to patient age and other factors. These differences are defined as follows [49]:

- **Newborn**: birth to first hospital discharge (see "Neonatal resuscitation in the delivery room", section on 'Resuscitation')
- **Infant**: younger than one year of age (including neonates, ie, less than 30 days of age, after first hospital discharge)
- **Child**: one year of age to the start of puberty (axillary hair in males and breast development in females)
- **Adult**: any patient either with signs of puberty or of older age (see "Adult basic life support (BLS) for health care providers")

Categories of BLS guidelines are determined by consensus due to lack of robust evidence determining the transition of BLS guidelines between newborn, infant, child, and adult physiologies [55,56].

BLS guidance for health care professionals for one rescuer is provided here and for two rescuers here [57].

The 2025 AHA cardiopulmonary resuscitation (CPR) and emergency cardiovascular care (ECC) guidelines reaffirm that the circulation-airway-breathing (C-A-B) sequence is still preferred for pediatric CPR [46,58]. Based upon the 2017 ILCOR treatment summary, conventional CPR has better outcomes than compression-only CPR (CO-CPR) and is preferred [51].

However, not all individual councils have adopted the 2020 pediatric BLS updates. As an example, for a single health care professional rescuer, A-B-C starting with the five stair step breaths are still advocated by the Resuscitation Council of the United Kingdom instead of C-A-B [59].

COVID-19 PATIENTS (SUSPECTED OR CONFIRMED)

The American Heart Association (AHA) has published guidance with algorithms for basic and advanced life support for children with suspected or confirmed coronavirus disease 2019 (COVID-19) [60-62]. Although not specifically designed for patients with other highly contagious and dangerous infections with airborne transmission, the approach would be expected to have similar benefits for health care professionals taking care of such patients.

For pediatric cardiac arrest, modifications to BLS include:

- Don personal protective equipment (PPE) prior to entering the patient room or scene (eg, cap, gown, N95 face mask, tight-fitting eye goggles or face shield, gloves) per local guidelines and availability.
- Limit personnel who are performing cardiopulmonary resuscitation (CPR); consider using a mechanical CPR device for adolescents who meet height and weight criteria.
- Provide rescue breathing using a bag-mask device with a high-efficiency particulate air (HEPA) filter and ensure a tight mask seal.

Preliminary evidence suggests that quality of BLS and rescuer fatigue is not significantly hampered by PPE. As an example, in a multicenter controlled trial involving 108 health care providers, neither chest compression quality (rate, depth, and release velocity) nor self-reported fatigue worsened to a significant degree in health care providers who performed BLS on pediatric manikins while wearing PPE, suggesting that the current pediatric BLS recommendation for chest compression providers to switch every two minutes need not be altered with PPE use [63].

BASIC LIFE SUPPORT APPROACH

Indications for BLS in critically ill or injured infants and children addressed in this topic are:

- Cardiac (pulseless) arrest
- Bradycardia (specifically heart rate <60/minute) with poor perfusion [42,45]

BLS guidance for health care professionals for one rescuer is provided here and for two rescuers here [57].

The sequence of key actions are [57]:

- **Verify scene safety** – Before beginning BLS, rescuers must ensure that the scene is safe for them and the victim (eg, removing the victim from a burning building or safely retrieving a drowning victim).
- **Determine unresponsiveness, get help, and activate emergency medical response system** – If the patient is unresponsive, then the rescuer should shout for nearby help and activate emergency medical services (EMS) via a mobile device (single rescuer outside of the hospital) or hospital system (eg, code button).

  Alternatively, with two or more health care providers, one rescuer continues to care for the victim and a second rescuer activates EMS and retrieves an automated external defibrillator (AED) and other emergency equipment (eg, code cart).

  If the patient is responsive, the rescuer should determine additional medical needs and need for activation of EMS based upon the patient's condition.

- **Assess breathing and pulse** – The rescuer should determine if the patient is breathing, only gasping, or not breathing while simultaneously checking for a pulse. Try to palpate a central pulse (brachial pulse in an infant; carotid or femoral pulse in a child) [57].

  This assessment guides further actions as follows [57]:

  - **No breathing or only gasping and no definite pulse within 10 seconds**:

    - **Single rescuer** – The approach depends upon whether the sudden collapse is witnessed or not:

      Unwitnessed collapse – For an unwitnessed, sudden collapse, the rescuer should start cardiopulmonary resuscitation (CPR; compressions-airway-breathing [C-A-B]) with a ratio of 30 compressions to 2 breaths. (See 'Chest compressions' below and 'Compression to ventilation ratio' below.)

      After about two minutes, if still alone, activate EMS and retrieve an AED (if not already done). Apply and activate the AED as soon as possible.

      Witnessed collapse – The rescuer should activate EMS (if not already done) and retrieve an AED or, for advanced life support (ALS) providers, a manual defibrillator. The single rescuer should then use the AED (or manual defibrillator for ALS providers) as soon as it is available. If the AED is not nearby or not available, start CPR while awaiting EMS arrival.

    - **Two or more rescuers** – The rescuer should begin CPR (C-A-B), starting with a ratio of 30 compressions to 2 breaths for a single rescuer and 15 compression to 2 breaths for two or more rescuers.

      In infants and children, the method of compressions and depth of compressions varies by age. (See 'Chest compressions' below.)

      Chest compressions in infants and children should always be accompanied by ventilation for infants and children who remain pulseless after the initial sequence of compressions. (See 'Compression to ventilation ratio' below.)

  Substantial evidence indicates that health care providers are often unable to quickly determine whether or not a pulse is present [64-67]. Consequently, when a pulse is not **definitely** identified within 10 seconds, CPR should be initiated.

  - **No normal breathing but pulse is present (same actions for single or multiple rescuers)**:

    - Start rescue breathing by providing one breath every two to three seconds (20 to 30 breaths per minute), which reflects a change from the 2015 guidelines suggested rate of every three to five seconds (12 to 20 breaths per minute).

    - Perform a pulse check for no longer than 10 seconds. Add chest compressions if the heart rate remains <60/minute with poor perfusion.

    - Activate EMS, if not already done.

    - Continue rescue breathing. Perform a pulse check every two minutes. If no pulse, begin CPR (C-A-B), starting with a ratio of 30 compressions to 2 breaths for a single rescuer and 15 compressions to 2 breaths for two or more rescuers.

In infants and children, the method of compressions and depth of compressions vary by age. (See 'Chest compressions' below.)

Chest compressions in infants and children should always be accompanied by ventilation for infants and children who remain pulseless after the initial sequence of compressions. (See 'Compression to ventilation ratio' below.)

- If an opioid overdose is suspected, give naloxone, if available.

- **Normal breathing and pulse is present (same action for single or multiple rescuers)**:
  - Monitor the patient until EMS arrives.

- **Initiate CPR** – The actions that constitute CPR are performing chest compressions, opening the airway, and providing ventilations, or C-A-B. (See 'Chest compressions' below and 'Breathing' below and "Basic airway management in children", section on 'Noninvasive relief of obstruction'.)

The sequence in which the actions of CPR for infants and children should be performed by health care providers is [57]:

- Initiate CPR in an infant or child who is unresponsive, has no normal breathing, and no definite pulse after 10 seconds.
- Start chest compressions **before** performing airway or breathing maneuvers (C-A-B).
- After 30 compressions (15 compressions if two rescuers), open the airway and give two breaths. (See 'Breathing' below.)
- If the heart rate is ≥60/minute after about two minutes of CPR, stop chest compressions and continue ventilation.
- Apply the AED or manual defibrillator:
  - **Single rescuer** – For a witnessed collapse, retrieve the AED or, for ALS providers, the manual defibrillator and use it as soon as possible. For an unwitnessed collapse, perform about two minutes of CPR and then activate EMS (if not already done) and retrieve the AED or, for ALS providers, the manual defibrillator.
  - **Two or more rescuers** – One rescuer initiates CPR while the other rescuer activates EMS and retrieves the AED or, for ALS providers, the manual defibrillator. The AED or manual defibrillator is then used as soon as it is available. (See "Technique of defibrillation and cardioversion in children (including automated external defibrillation)", section on 'Procedure' and "Technique of defibrillation and cardioversion in children (including automated external defibrillation)", section on 'Automated external defibrillator use in infants and children'.)
- Proceed based upon AED analysis as follows:
  - **Shockable rhythm** – Give one shock and resume CPR immediately for about two minutes (until prompted by the AED to allow a rhythm check). Continue CPR with pulse check and AED rhythm check every two minutes until ALS providers take over or the victim starts to move.
  - **Non-shockable rhythm** – Resume CPR immediately for about two minutes (until prompted by the AED to allow a rhythm check). Continue CPR with pulse check and AED rhythm check every two

minutes until ALS providers take over or the victim starts to move.

The algorithms for a single and two or more rescuers are designed so that CPR is performed for approximately two minutes (five cycles for single rescuer and 10 cycles for two or more rescuers) before using an AED in a patient with an unwitnessed arrest. This approach is based upon limited evidence in adults that, even for prolonged arrest from ventricular fibrillation (VF), an initial period of CPR improves the likelihood of successful defibrillation. (See "Adult basic life support (BLS) for health care providers", section on 'Phases of resuscitation'.)

CHEST COMPRESSIONS

The 2025 American Heart Association (AHA) cardiopulmonary resuscitation (CPR) and emergency cardiovascular care (ECC) guidelines reaffirm prior treatment recommendation and continue to emphasize the importance of proper technique when performing chest compressions [50,54,57]. Chest compressions should be performed over the lower half of the sternum. **Avoid** compression of the xiphoid process that can cause trauma to the liver, spleen, or stomach.

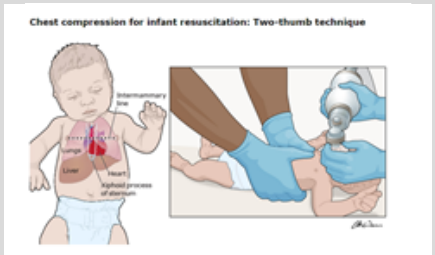
The effectiveness of chest compressions can be maximized by attention to the following essential elements [57]:

- The chest should be depressed at least one-third of its anterior-posterior diameter with each compression [68-71]:
  - **Infants**: approximately 4 cm (1.5 inches)
  - **Children**: 5 cm (2 inches)
  - **Adolescents**: 5 to 6 cm, but should not exceed 6 cm (2.4 inches) per adult recommendations
- The optimum rate of compressions is approximately 100 to 120 per minute [72,73]. Each compression and decompression phase should be of equal duration.
- The sternum should return briefly to its normal position at the end of each compression, allowing the chest to recoil fully [74,75].
- A smooth compression-decompression rhythm, with minimal interruption, should be developed [30,76].
- Perform chest compressions on a firm surface [77-83]. During in-hospital cardiopulmonary arrest (IHCA), use a backboard to improve chest compression depth [77,79,80,84-87], and when available, activate the "CPR mode" of the bed to increase mattress stiffness [77,88-90].

Use of coaching, audio-visual feedback, and frequent training improves provider adherence to these recommendations. (See 'CPR coaching, feedback devices, and refresher training' below.)

Infants — For infants undergoing two-rescuer CPR, perform compressions using the two-thumb-encircling hands technique or heel of one hand (one-hand) technique [50,57]:

- **Two-thumb-encircling technique** – The thorax is encircled with both hands, and chest compressions are performed with the thumbs (figure 1). The thumbs compress the lower half of the sternum, avoiding the xiphoid process, while the hands are spread around the thorax [50,57].



- **One-hand technique** – Chest compressions are performed over the sternum with the heel of one hand (figure 2). For infants undergoing single rescuer CPR, the one-hand technique is used rather than the two-thumb-encircling technique because it permits a more rapid transition from compressions to ventilation.

Chest compression for infant resuscitation: Two-thumb technique

Figure 1 - larger image below

Limited evidence from multicenter prospective registry studies suggest that the quality of compressions are similar for the two-thumb encircling and one-hand techniques in infants [91,92].

The two-finger technique is no longer recommended because, compared with the two-thumb-encircling or one-hand technique, it is less effective in achieving the proper depth based upon infant manikin studies and experience reported from a multicenter prospective observational registry study [91,93].

Children — For children, chest compressions should be performed over the lower half of the sternum with either the heel of one hand or with two hands (picture 1 and picture 2).

No outcome studies have compared these two techniques in children with cardiac arrest [57]. In pediatric manikin studies, the two-hand technique has been associated with improved compression depth [94], improved compression force [95], and less rescuer fatigue [96].

Patients with mechanical circulatory support — For patients with durable mechanical circulatory support (eg, left ventricular assist devices [LVADs]) who have acutely impaired perfusion caused by cardiac arrest and who are outside the immediate peri-device implantation period, chest compressions are indicated [54,97,98].The resuscitation team should limit delays in performing chest compressions while simultaneously checking for device-related reversible causes of the acute event. An algorithmic approach to these patients is available here.

Durable mechanical circulatory support, primarily LVADs have become a more common method to manage children with advanced heart failure, primarily as a bridge to heart transplant (see "Heart failure in children: Management", section on 'Mechanical circulatory support'). Limited evidence from case reports and case series in children and adults with LVADs and impaired perfusion or cardiac arrest suggests that cardiac compressions performed outside the immediate peri-device implantation period are unlikely to dislodge an LVAD [97,98]. Furthermore, although some reports have suggested that intact neurologic survival is worse in patients with LVADs who receive chest compressions, these studies are likely confounded by misclassification of the true circulatory status of the patient [54]. Thus, for patients with LVADs who have acutely impaired perfusion caused by cardiac arrest, cardiac compressions are suggested while troubleshooting device-related causes of acutely impaired perfusion.

CPR coaching, feedback devices, and refresher training — For pediatric CPR, coaching, real-time feedback devices, and refresher training appear to improve the quality of chest compressions during pediatric resuscitation although evidence of improved clinical outcomes with the use of CPR coaching or feedback devices in children with cardiac arrest is lacking [99-110]:

- Coaching – Specific tasks of the CPR coach include [111]:
  - Coordinates the start of CPR
  - States the midrange compression and ventilation rate targets
  - Provides device and visual feedback on quality of chest compressions to enhance performance

- Minimizes pauses during CPR with the goal of maintaining a cardiac compression fraction (CCF; proportion of time during cardiac arrest during which chest compressions are administered) at 80 percent

During pediatric resuscitation, the CPR coach may also act in the monitor/defibrillator role.

Compared with team performance without a CPR coach, the inclusion of a trained CPR coach during simulated pediatric cardiac arrest has been associated with decreased team member mental workload [107] and better team performance (improved verbalization before pauses in CPR, decreased pause duration, shorter pauses during intubation, and better coordination of key tasks during pauses) [108,112].

- Feedback devices – Mechanical feedback devices (audio and audio-visual) plus integration of a trained CPR coach into resuscitation teams has improved chest compression quality in randomized, simulation-based clinical trials [99-103,105,109,110]. Mechanical feedback device use alone has been associated with an increased likelihood of delivery of chest compressions compliant with excellent CPR (rate, depth, and CCF) during in-hospital cardiac arrest [106].

- Refresher training – Refresher CPR simulation training ("just-in-time" CPR training with use of a visual feedback device) provided prior to trial CPR performance combined with visual feedback during trial performance may improve CPR quality during simulated pediatric cardiac arrest more than either refresher training or visual feedback alone, especially when performed on a regular basis [100]. For example, CPR simulation training with real-time feedback occurring monthly significantly improved the proportion of health care providers able to perform "excellent CPR" (defined as 90 percent guideline compliance for compression depth, rate, and recoil) on an infant manikin compared with providers in the control group receiving only annual BLS training (72 versus 19.5 percent, respectively) [104]. In a prospective observational study, a high-frequency training program in pediatric CPR attended by ED providers in a children's hospital over a 15-month period resulted in improved performance of CPR skills during simulated and actual pediatric cardiac arrest [113].

COMPRESSION TO VENTILATION RATIO

No advanced airway — For infants and children who remain pulseless after the initial sequence of compressions, chest compressions should always be accompanied by ventilation [13,48,49,64,114,115]. However, every effort should be made to avoid excessive ventilation and to limit interruptions of chest compressions to less than 10 seconds [50,57].

Experimental evidence in animals indicates that coronary artery perfusion pressure declines with interruptions in chest compressions [116,117]. Observational reports suggest that pauses in chest compressions negatively impact return of spontaneous circulation and survival after CPR [118,119].

Compression to ventilation ratios of 30 to 2 and 15 to 2 are recommended to minimize interruption and for ease of teaching and retention [64,114].

- **Single rescuer**: Two ventilations should be delivered during a short pause at the end of every 30th compression

- **Two rescuers**: Two ventilations should be delivered at the end of every 15th compression

Advanced airway — Once the trachea is intubated, chest compressions and ventilations can be performed independently. For infants and children, chest compressions are delivered at a rate of 100 to 120 per minute

without pauses, and ventilations are given at a rate of 20 to 30 breaths per minute (one breath every two to three seconds). (See "Pediatric advanced life support (PALS)", section on 'Advanced airway placement'.)

Conventional versus compression-only CPR — In children, CPR administered by health care providers consists of both chest compressions and ventilations (ie, conventional CPR). This is because hypoxic/asphyxial arrest is the most common type of cardiac arrest in pediatric patients and therefore providing adequate ventilation is an essential part of resuscitation. (See 'Epidemiology and survival' above.)

Compression-only CPR (CO-CPR) plays a very limited role in pediatric resuscitation [57,120-124]. CO-CPR is warranted only for the rare infant or child with cardiac arrest who cannot be ventilated (eg, due to complete airway obstruction or severe lung disease). In this setting, CO-CPR should be provided while oxygenation and ventilation are established.

This differs from adult BLS guidelines, which encourage use of CO-CPR more broadly when personnel to perform conventional CPR are not available (eg, a sole lay rescuer who is reluctant to deliver breaths). (See "Adult basic life support (BLS) for health care providers", section on 'Compression-only CPR'.)

Data on outcomes with conventional CPR versus CO-CPR come from observational studies of children with OHCA who received bystander-administered CPR [22,120,125,126]. In these studies, children who received both chest compressions and rescue breaths (ie, conventional CPR) generally had better outcomes at one month (ie, survival with favorable neurologic outcome) compared with those who received CO-CPR or no CPR, though the estimates varied widely among the studies (conventional CPR: 13 to 26 percent; CO-CPR: 9 to 16 percent; no bystander CPR: 2 to 7.5 percent) [22,120,125,126].

AIRWAY

The airway should be opened with a head tilt-chin lift maneuver unless a cervical spine injury is suspected. In the setting of trauma in which cervical spine injury is suspected, use a jaw thrust maneuver without head tilt; however, if this action is unsuccessful, use the head tilt-chin lift maneuver. (See "Basic airway management in children", section on 'Noninvasive relief of obstruction'.)

BREATHING

Ventilations can be provided with mouth-to-mouth, mouth-to-nose, or with a bag and mask (see "Basic airway management in children", section on 'Assisted ventilation'). Bag-mask ventilation is often sufficient to achieve adequate ventilation during cardiopulmonary resuscitation (CPR) [54]. During CPR, bag-mask ventilations with 100 percent inspired oxygen are provided. If adequate ventilation cannot be achieved with bag-mask ventilation, or the resuscitation is prolonged, an advanced airway (endotracheal tube or supraglottic device) may be required. Bag-mask ventilation is a reasonable alternative to an advanced airway (endotracheal intubation or supraglottic airway) in the prehospital management of pediatric cardiac arrest.

Basic and advanced airway management during prehospital pediatric care is discussed in greater detail separately. (See "Pediatric considerations in prehospital care", section on 'Basic airway management' and "Pediatric considerations in prehospital care", section on 'Advanced airway management'.)

Evidence in adults and animals suggest that hyperventilation is associated with increased intrathoracic pressure and decreased coronary and cerebral perfusion (see "Adult basic life support (BLS) for health care providers", section on 'Ventilations'). These data are the basis for the following recommendations:

- Each rescue breath should be delivered over one second.
- The volume of each breath should be sufficient to see the chest wall rise.
- An infant or child with a heart rate ≥60/minute without normal breathing should receive one breath every two to three seconds (20 to 30 breaths per minute) [50].
- Infants and children who require chest compressions should receive two breaths per 30 chest compressions for a lone rescuer and two breaths per 15 chest compressions for two rescuers. (See 'Compression to ventilation ratio' above.)
- Intubated infants and children should be ventilated at a rate of 20 to 30 breaths per minute (one breath every two to three seconds) with a goal of 30 breaths per minute in infants and 20 to 25 breaths per minute in children without any interruption of chest compressions [50,54,120]. (See 'Compression to ventilation ratio' above.)

AUTOMATED EXTERNAL DEFIBRILLATOR

An automated external defibrillator (AED) is a portable device that identifies "shockable rhythms" that should be treated with defibrillation. It then instructs the operator how to use the device to deliver a standard shock to the patient. The device also identifies "non-shockable rhythms" and advises no shock followed by a prompt to resume cardiopulmonary resuscitation (CPR). (See "Automated external defibrillators" and "Technique of defibrillation and cardioversion in children (including automated external defibrillation)", section on 'Automated external defibrillator use in infants and children'.)

The 2025 American Heart Association (AHA) and American Academy of Pediatrics guidelines for CPR and Emergency Cardiovascular Care (ECC) reaffirm the following for infants and children with cardiac arrest [50,57]:

- **Witnessed arrest** – An AED should be used as soon as possible if a manual defibrillator is not available [50,57]. CPR should be performed until the AED (or manual defibrillator, if available) is ready to deliver a shock [121-125]. A single shock followed by immediate chest compressions is recommended for infants and children with a shockable rhythm. The initial manual defibrillation dose is 2 J/kg (up to the adult maximum dose) with timing of additional defibrillation and escalation of defibrillation doses as indicated in the algorithm and discussed separately. (See "Pediatric advanced life support (PALS)", section on 'Shockable rhythm' and "Technique of defibrillation and cardioversion in children (including automated external defibrillation)", section on 'Methods: Manual defibrillator use'.)
- **Unwitnessed arrest** – The algorithms for a single rescuer and two or more rescuers are designed so that CPR is performed for approximately two minutes (five cycles for single rescuer and 10 cycles for two or more rescuers) before using an AED. (See 'Basic life support approach' above.)
- **Age <8 years** – An AED with a pediatric dose attenuating system should be used whenever possible [2,3,31,126-133]. However, if a manual defibrillator or an AED with a pediatric dose attenuating system is not available, use of an AED without a dose attenuator is advised [13,115,128,129,131,134]. When affixing the self-adhering pads, use either anterior-lateral placement or anterior-posterior placement to maintain good separation between the pads [135,136]. (See "Technique of defibrillation and cardioversion in children (including automated external defibrillation)", section on 'Automated external defibrillator use in infants and children'.)

In observational series, up to 19 percent of infants and children in cardiac arrest outside of a hospital (OHCA) had a shockable rhythm as the initial rhythm. Thus, a substantial number of children in cardiac arrest might benefit from early defibrillation [1,7,8,29,137]. Limited evidence suggests that AEDs can be appropriately and safely used for infants and children [126-130,138].

The AED is designed to be used by untrained bystanders and is increasingly available in public locations such as airports, athletic events, shopping malls, and the workplace. The American Academy of Pediatrics (AAP) recommends that lay rescuer AED programs (such as in schools) be implemented as part of comprehensive emergency response plans, rather than as programs focused on a single piece of equipment.

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Basic and advanced cardiac life support in children".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "Patient education: CPR for children (The Basics)" and "Patient education: CPR for infants (The Basics)")

SUMMARY AND RECOMMENDATIONS

- **Prompt recognition** – Pediatric cardiac arrest most often results from progressive tissue hypoxia and acidosis due to respiratory failure and/or shock. However, sudden arrest from a shockable rhythm (ie, ventricular fibrillation [VF] or pulseless ventricular tachycardia [pVT]) also occurs frequently. Survival rates for children with cardiac arrest are improved by prompt recognition and initiation of cardiopulmonary resuscitation (CPR) as summarized in the basic life support (BLS) algorithms and stepwise approach below. (See 'Epidemiology and survival' above.)

- **BLS algorithms** – BLS guidance for health care professionals for one rescuer is provided here and for two rescuers here. (See 'Basic life support approach' above.).

  Indications for CPR include:

  - Cardiac (pulseless) arrest (algorithm 1)

- Bradycardia (heart rate <60/minute) with a pulse and poor perfusion (algorithm 2)

Guidance for the resuscitation of newborns is reviewed separately. (See "Neonatal resuscitation in the delivery room".)

- **Assess breathing and check for pulse** – The first step is to determine whether the patient is breathing while simultaneously checking for a pulse. For infants and children with signs of cardiac arrest, check for a pulse (for infants, check the brachial pulse; for children, check the femoral or carotid pulse). The pulse check should last no longer than 10 seconds. After initiating CPR, perform subsequent pulse checks approximately every two minutes.

- **Start CPR** – Start CPR if the child has no or ineffective breathing and no definite pulse within 10 seconds. CPR consists of both chest compressions and ventilation in most cases. The exception is the rare patient with complete airway obstruction or severe lung disease in whom ventilation may not be possible initially; compression-only CPR is appropriate for such patients until the airway is established. (See 'Conventional versus compression-only CPR' above.)

  Usual practice is to start CPR with compressions before breathing (ie, compressions-airway-breathing [C-A-B]), although it is also acceptable to start with breathing (ie, airway-breathing-compressions [A-B-C]). (See 'Basic life support approach' above.)

- **Chest compressions**

  - **Depth of compressions** – Each chest compression should depress the chest by a minimum of one-third of its anterior-posterior diameter, at a rate of about 100 to 120 compressions per minute. The technique and required depth of compressions depends upon patient age (see 'Chest compressions' above):

    - **Infants (<1 year)** – For infants undergoing single-rescuer CPR, perform compressions using the heel of one hand (figure 3). For infants undergoing two-rescuer CPR, perform compressions using the two-thumb encircling hands (figure 1) or one-hand technique (see 'Infants' above). The required depth is approximately 4 cm (1.5 inches). If the rescuer is unable to encircle the chest with their hands, it is reasonable to use the heel of one hand. (See 'Infants' above.)

    - **Children (≥1 year)** – In children, perform chest compressions using either a one-hand or two-hand technique. The required depth is 5 cm (2 inches). (See 'Children' above.)

    - **Adolescents** – In adolescents, perform chest compressions with two hands per adult recommendations. The required depth is 5 to 6 cm but should not exceed 6 cm (2.4 inches). (See "Adult basic life support (BLS) for health care providers", section on 'Chest compressions'.)

  Chest compressions should be performed on a firm surface (eg, in health care facilities, use a backboard or activate "CPR mode" on the bed). The chest should fully recoil at the end of each compression. Interruptions in chest compressions should be minimal (<10 seconds). The use of CPR feedback devices (eg, metronome, visual feedback, and coaching) may help to optimize the rate of compressions.

- **Airway** - Open the airway (see "Basic airway management in children"):

  - No concern for cervical spine injury – Use a head tilt-chin lift maneuver (figure 4)
  - Concern for cervical spine injury (eg, trauma) – Use a jaw thrust maneuver (figure 5). If this is unsuccessful, carefully use the head tilt-chin lift maneuver.

- **Breathing** – Deliver each breath over one second with enough volume to see the chest wall rise. Avoid excessive ventilation. Bag-mask ventilation (BMV) usually provides adequate ventilation during CPR. If adequate ventilation cannot be achieved with BMV or the resuscitation is prolonged, an advanced airway (endotracheal tube or supraglottic device) may be required. (See 'Breathing' above.)

  - **Rescue breathing in children without cardiac arrest** – For infants and children who are not in cardiac arrest (ie, heart rate ≥60/minute) but who have absent or inadequate respiratory effort, provide rescue breaths at a rate of one breath every two to three seconds (20 to 30 breaths per minute). (See 'Breathing' above.)

  - **Compression-ventilation ratio**: The compression-ventilation ratio for children depends upon the number of rescuers and whether an advanced airway is in place (see 'Compression to ventilation ratio' above):

    - **Single rescuer**: 30 compressions followed by two breaths (30:2)

    - **Two or more rescuers**: 15 compressions followed by two breaths (15:2)

    - **Advanced airway in place**: 20 to 30 breaths per minute (one breath every two to three seconds) with a goal of 30 breaths per minute in infants and 20 to 25 breaths per minute in children without any interruption of chest compressions

- **Administer naloxone** – If an opioid overdose is suspected, give naloxone, if available (table 2). (See 'Basic life support approach' above and "Opioid intoxication in children and adolescents", section on 'Dosing and administration'.)

- **Check rhythm and, if shockable, perform defibrillation** – All children in cardiac arrest require assessment for a shockable rhythm using an automated external defibrillator (AED) or, for advanced life support (ALS) providers, a manual defibrillator. (See 'Automated external defibrillator' above.)

  When affixing the self-adhering pads, use either anterior-posterior placement or anterior-lateral placement to maintain good separation between the pads. For children <8 years of age, use an AED with a pediatric dose attenuating system, if available. However, if neither a dose-attenuating system nor a manual defibrillator is available, it is acceptable to use an AED without a dose attenuator.

  The sequence of applying the defibrillator depends on whether the arrest is witnessed or unwitnessed:

  - **Witnessed arrest** – For a witnessed arrest, the AED (or, for ALS providers, manual defibrillator) should be applied as soon as possible. Perform CPR until the AED (or manual defibrillator) is ready to deliver a shock; immediately start chest compressions after the shock is delivered.

  - **Unwitnessed arrest** – For an unwitnessed arrest, perform CPR for approximately two minutes (five cycles for single rescuer and 10 cycles for two or more rescuers) before using an AED or manual defibrillator.

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Topic 6384 Version 53.0

GRAPHICS

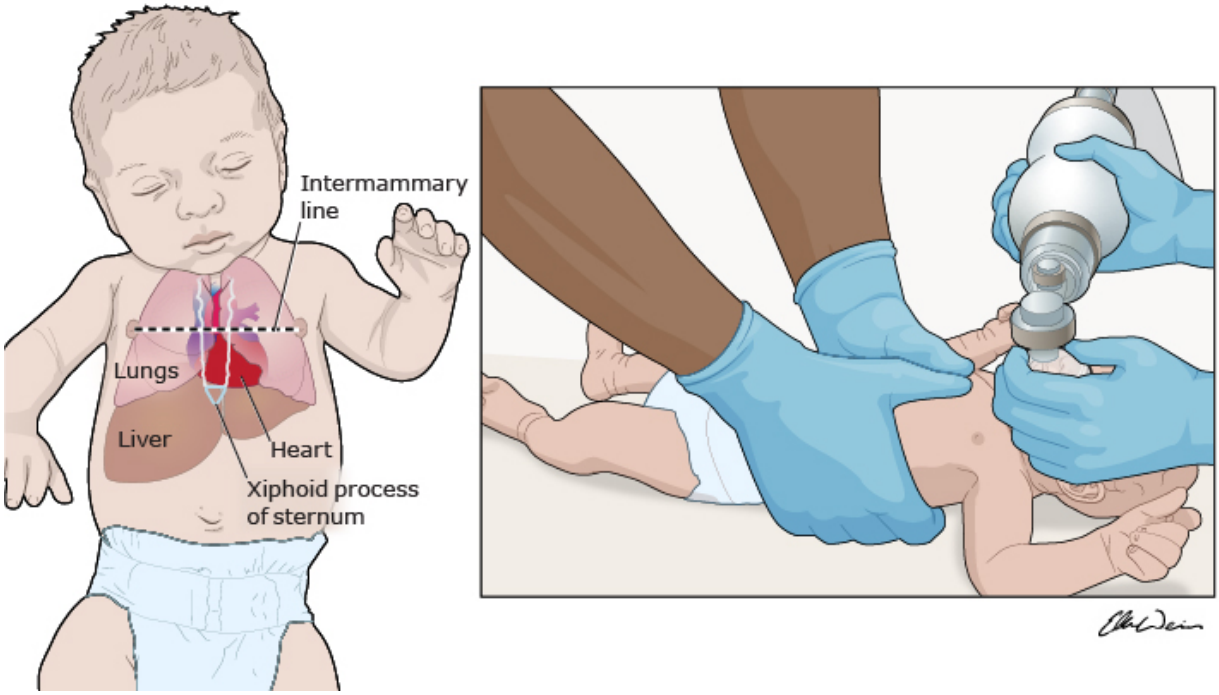
Table 1: **Drug- and toxin-induced electrocardiographic abnormalities**

Bradycardia/AV blockade	Supraventricular tachycardia	Ventricular tachycardia	QRS and QT interval prolongation
Beta blockers	Sympathomimetics	Sympathomimetics	Antidepressants
Calcium channel blockers	Amphetamines	Cocaine	Citalopram
Cardiac glycosides	Cocaine	Amphetamines	Escitalopram
Digoxin	Theophylline	Theophylline	Bupropion
Digitoxin	Caffeine	Antidepressants	Venlafaxine
Red squill	Methylphenidate	TCAs	Desvenlafaxine
Digitalis lanata	Ephedrine	Antipsychotics	Tricyclic antidepressants
Digitalis purpurea	Pseudoephedrine	Phenothiazines	Antipsychotics
Bufotenin	Albuterol	Chlorinated hydrocarbons	Antihistamines
Oleander	Dobutamine	Chloral hydrate	Diphenhydramine
Alpha-adrenergic agonists	Epinephrine	Solvents	Astemizole
Phenylpropanolamine	Dopamine	Fluoride	Terfenadine
Clonidine	Anticholinergics	Cardiac glycosides	Antiarrhythmics
Imidazolines	Antihistamines	Potassium	Quinidine
Cholinergics	TCAs		Disopyramide
Organophosphates	Phenothiazines		Procainamide
Carbamates	Clozapine		Propafenone
Opioids	Atropine		Flecainide, encainide
Sedative-hypnotics	Scopolamine		Amiodarone
Magnesium	Thyroid hormone		Calcium channel blockers (rare)
	Cellular asphyxiants		Beta blockers (rare)
	Carbon monoxide		Ondansetron
	Drug withdrawal states		Propoxyphene
			Organophosphate insecticides
			Antimicrobials
			Amantadine
			Azithromycin
			Chloroquine
			Erythromycin
			Pentamidine
			Quinine
			Quinolones (eg, ciprofloxacin)

Arsenic
Thallium
Fluoride
Citrate
Lithium

Graphic 66773 Version 10.0

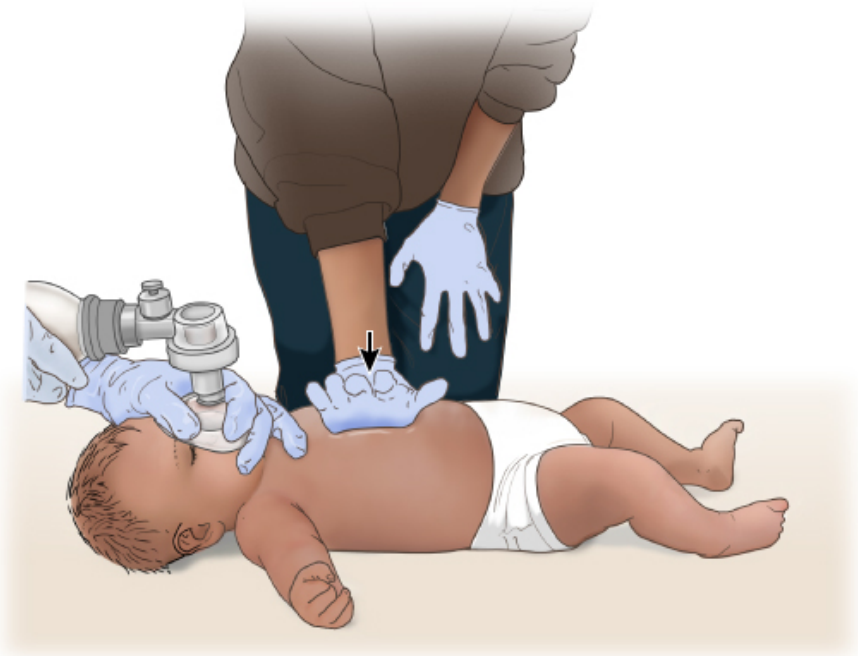
Figure 1: **Chest compression for infant resuscitation: Two-thumb technique**



The thorax is encircled with the hands and cardiac compressions are performed with both thumbs. The compression site is approximately one finger's breadth below the intermammary line. The area over the xiphoid process should be avoided to prevent injury to the liver, spleen, or stomach.

Graphic 77050 Version 4.0

Figure 2: **One-hand cardiac compressions in an infant**



Chest compressions are performed over the sternum with the heel of one hand.

Graphic 150271 Version 1.0

Picture 1: **One-handed chest compressions**

For children (from one year until the start of puberty), chest compressions may be performed with the heel of one hand placed over the lower half of the sternum.

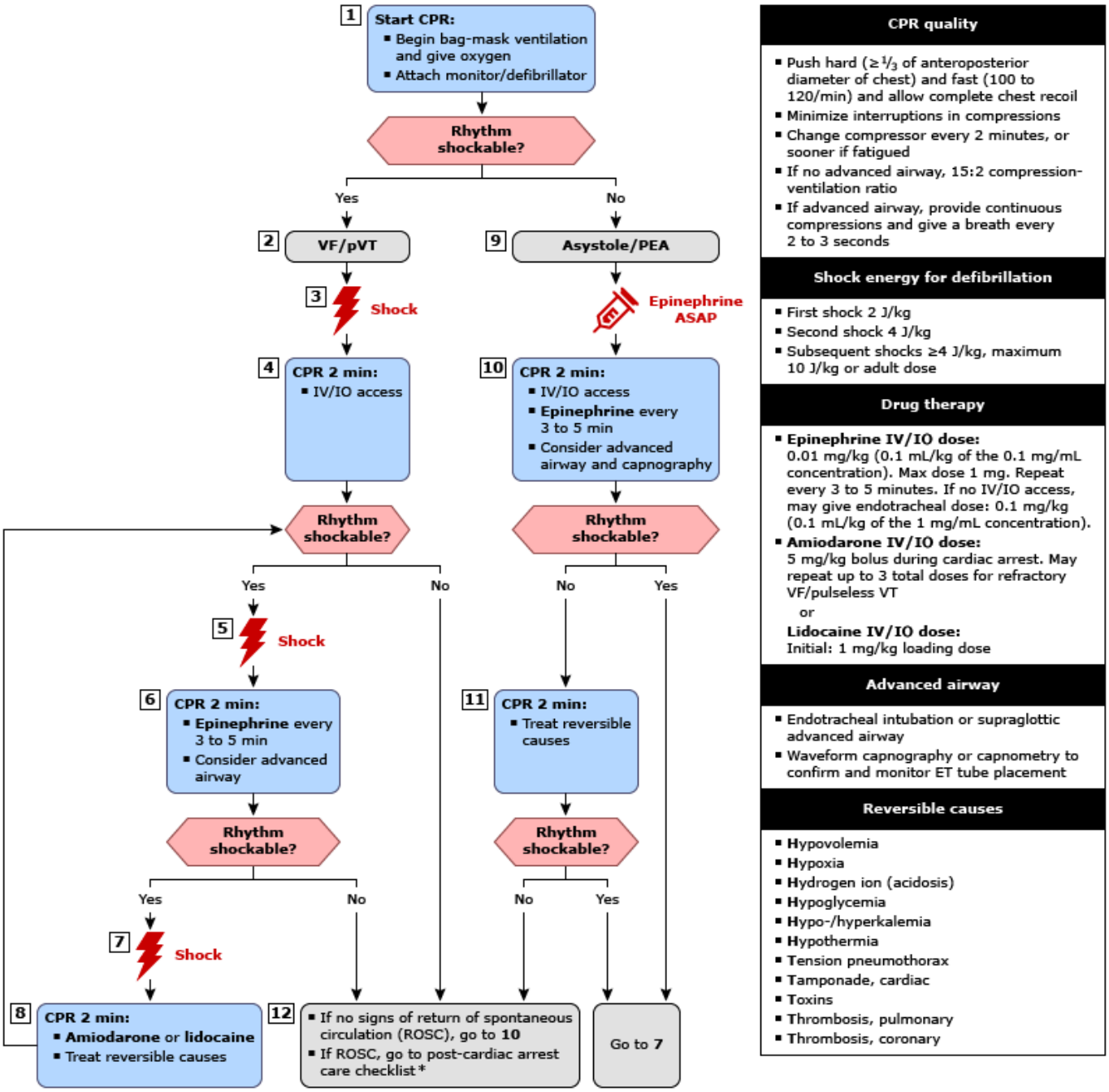
Graphic 62171 Version 2.0

Picture 2: **Two-handed chest compressions**

For children (from one year until the start of puberty), chest compressions may be performed with two hands placed over the lower half of the sternum.

Graphic 59710 Version 2.0

Algorithm 1: **Pediatric cardiac arrest algorithm 2020 update**



Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Pediatric Cardiac Arrest Algorithm is now available. This algorithm will be updated within UpToDate in the future.

CPR: cardiopulmonary resuscitation; ET: endotracheal; IO: intraosseous; IV: intravenous; PEA: pulseless electrical activity; pVT: pulseless ventricular tachycardia; ROSC: return of spontaneous circulation; VF: ventricular fibrillation.

* If ROSC, go to post-cardiac arrest care checklist. The post-cardiac arrest care checklist can be found here.

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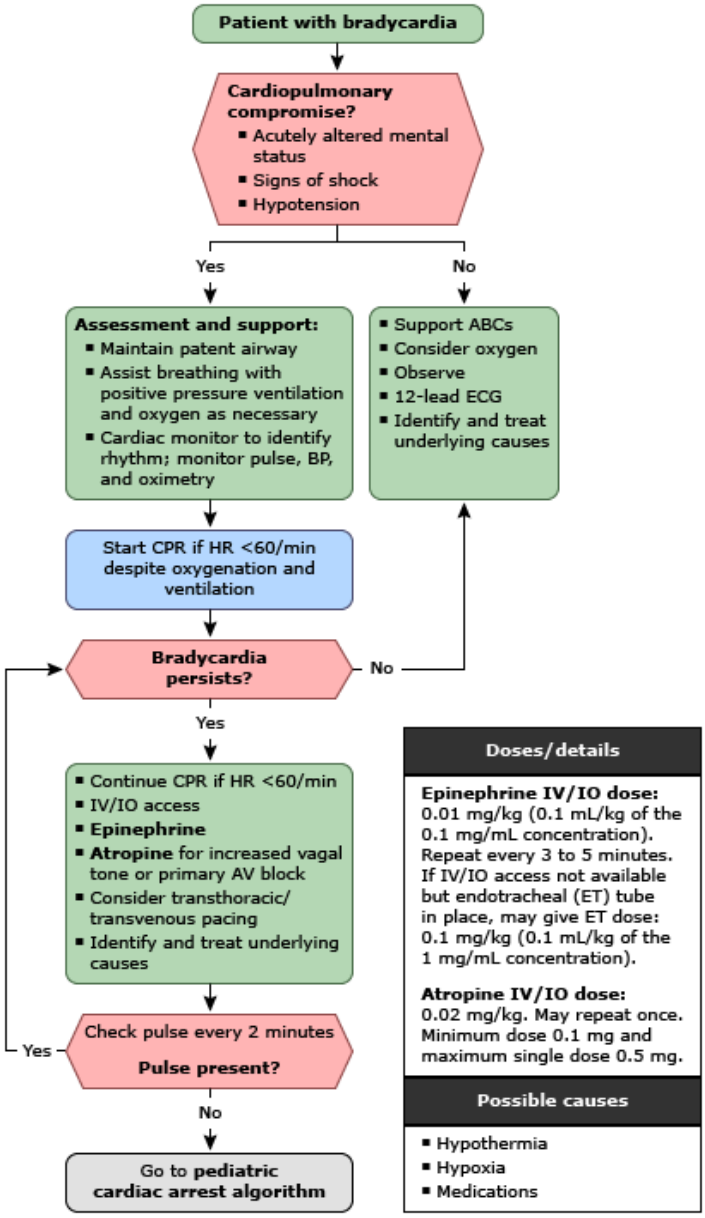
Graphic 129942 Version 12.0

Algorithm 2: **Pediatric bradycardia with a pulse 2020 update**



Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Pediatric Bradycardia with a Pulse Algorithm is now available. This algorithm will be updated within UpToDate in the future.

ABCs: airway, breathing, circulation; AV: atrioventricular; BP: blood pressure; CPR: cardiopulmonary resuscitation; ECG: electrocardiogram; ET: endotracheal; HR: heart rate; IO: intraosseous; IV: intravenous.

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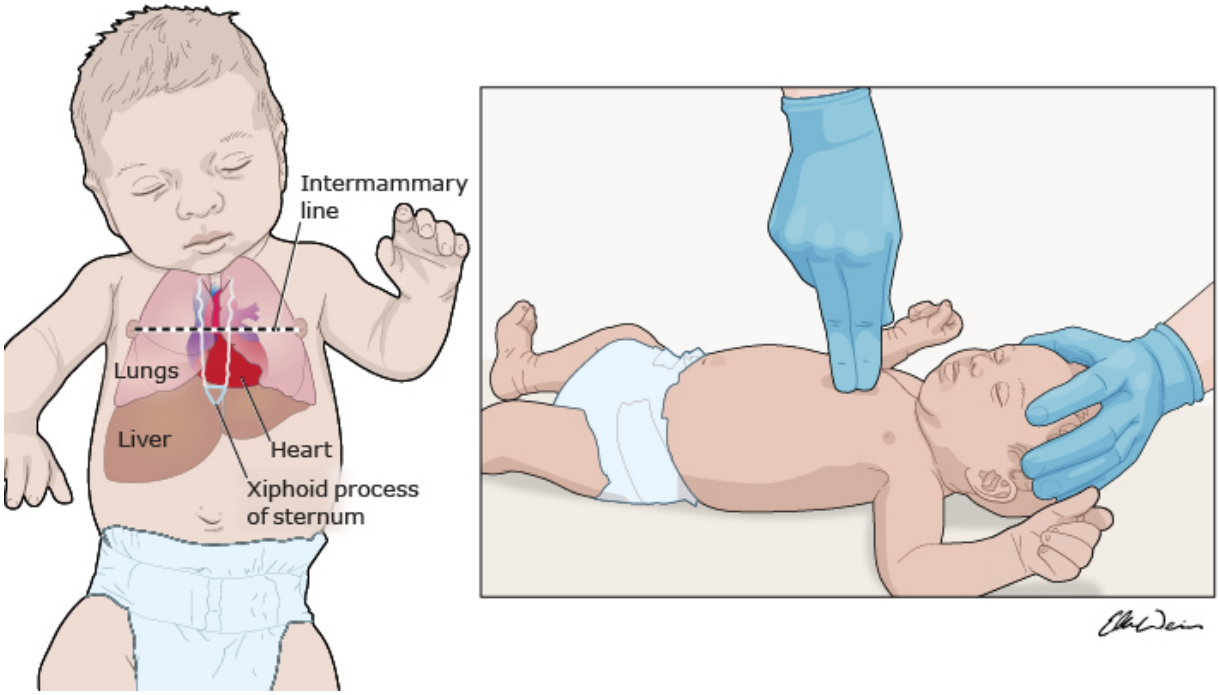
Graphic 129941 Version 11.0

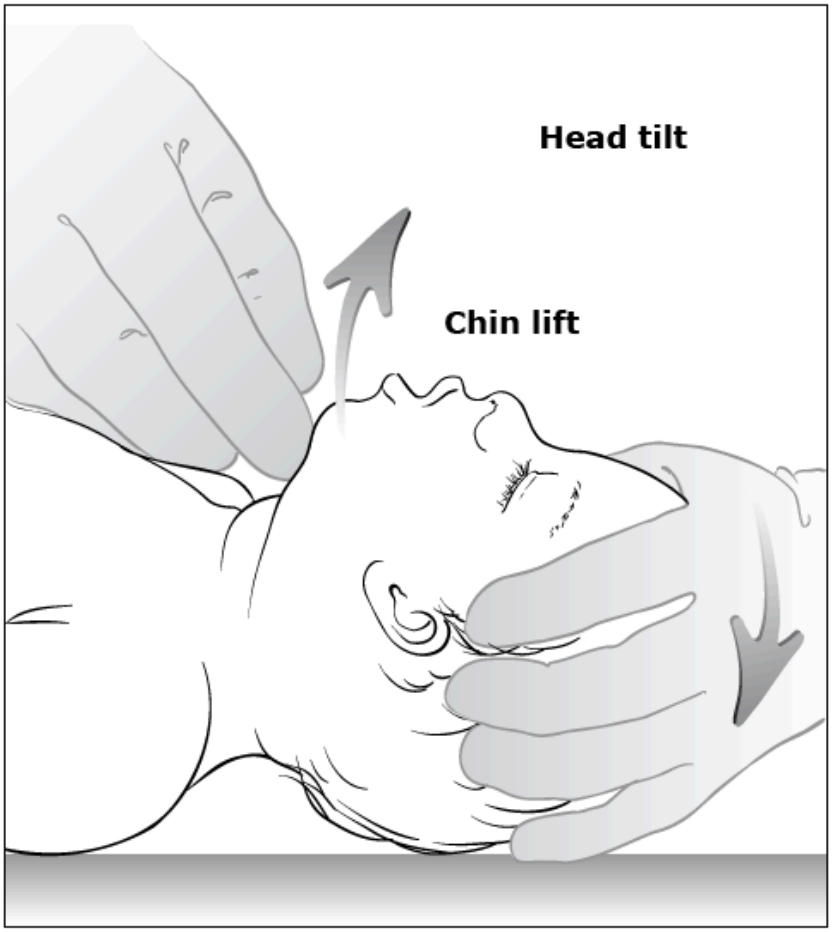
Figure 3: **Chest compression for infant resuscitation: Two fingers technique (Single rescuer)**



Chest compressions for infants (under one year) may be performed with two fingers placed on the sternum just below the nipples. This picture shows the site of compressions. When compressions are performed the two fingers used should be perpendicular to the chest and straight.

Graphic 73038 Version 6.0

Figure 4: **Head tilt-chin lift in an infant**



Head tilt-chin lift maneuver. Note that the fingers should not extend over the submental area, as this may produce airway obstruction.

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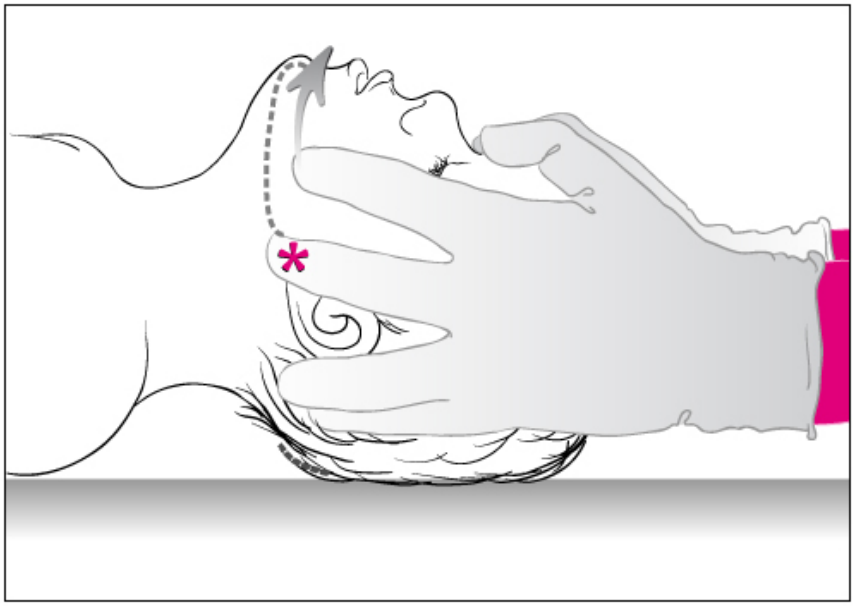
Graphic 139464 Version 1.0

Figure 5: **Jaw thrust maneuver in a child**



With the jaw thrust maneuver, the clinician can open the airway and limit neck motion. While maintaining the head and neck in a neutral position as shown, hook the middle finger of both hands under the angle of the jaw (asterisk) while providing countertraction with both thumbs on the forehead. Lift the mandible upward (arrow) to open the mouth and move the tongue anteriorly. The clinician can place their elbows on the surface above the patient's head to brace themselves and maintain the jaw thrust more comfortably.

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Graphic 139465 Version 1.0

Table 2: **Opioid intoxication in children and adolescents: Rapid overview of emergency management**

To obtain emergency consultation with a medical toxicologist, in the United States, call 1-800-222-1222 for the nearest regional poison control center. Contact information for poison control centers around the world is available at the WHO website and in the UpToDate topic on regional poison control centers (society guideline links).
Clinical features*
Depressed mental status ranging from lethargy to coma
Miotic pupils (often pinpoint)¶
Depressed respirations including apnea
Bradycardia and hypotension
Decreased to absent bowel sounds
Track marks (adolescents with IV drug abuse)
Ancillary studiesΔ
Blood glucose and pulse oximetry in patients with coma and depressed respirations
Serum ethanol level, especially in adolescents with recreational or suicidal overdose
Serum acetaminophen if suicidal intent or ingestion of combination medications (eg, acetaminophen with oxycodone)
Electrocardiogram in patients intoxicated with methadone (prolonged QTc) or propoxyphene (prolonged QRS)
Chest radiograph in patients with persistent respiratory findings that suggest pulmonary aspiration or noncardiogenic pulmonary edema
Urinalysis, serum electrolytes, blood urea nitrogen, creatinine, and creatine kinase in patients with prolonged immobilization or muscular rigidity at risk for rhabdomyolysis
Rapid urine pregnancy test in postmenarchal girls
Rapid urine testing for opioids in young children and infants with coma from unknown cause or in whom intentional poisoning by a caretaker is suspected
Treatment
Support airway and breathing using bag-mask ventilation with 100% inspired oxygen until naloxone is administered.
Give naloxone for deep coma and respiratory depression as follows◇: ▪ Children <20 kg: 0.1 mg/kg IV or IO (maximum 2 mg per dose) **except neonates**§. ▪ Children over 20 kg: 2 mg IV or IO. ▪ Adolescents with suspected opioid dependence and spontaneous ventilation: 0.04 to 0.08 mg per dose repeated every few minutes and titrated to patient response. ▪ If no effect, repeat the naloxone dose every one to two minutes to a maximum total dose of 10 mg. ▪ Patients with recurrent toxicity may receive additional bolus doses or a continuous naloxone infusion. Begin the infusion rate at 2/3 of the total dose of naloxone needed to restore breathing, delivered every hour. • If the patient develops respiratory depression despite the naloxone infusion (this may happen 20 to 30 minutes after starting infusion) administer a naloxone bolus (using half the original effective bolus dose) and repeat if necessary until adequate ventilation returns, then increase the infusion rate. • If the patient develops signs of opioid withdrawal, stop the infusion. If respiratory depression returns, start the infusion at half the original rate.
Give activated charcoal (1 gram/kg orally or by nasogastric tube, maximum dose 50 grams) to alert young children and adolescents who present within one hour of oral overdose¥.
In cases of suspected child abuse, consult an experienced children protection team (if available) and report suspicion to the appropriate governmental agency.
Consult psychiatry for patients with substance abuse or suicidal intent.

AC: activated charcoal; ET: endotracheal tube; IM: intramuscular; IO: intraosseous.

* The diagnosis of opioid intoxication is based upon clinical features and response to antidotal therapy.

¶ Small pupils are not always present in children and adolescents with opioid intoxication.

Δ Other ancillary studies may be indicated in selected patients in whom trauma, infection, or other etiologies besides poisoning are suspected. Refer to UpToDate topics on stupor and coma in children and pediatric occult toxic exposures.

◇ Naloxone may also be given intramuscularly or intratracheally via endotracheal tube (ET) in children or adolescents with poor IV access. If an endotracheal tube is present, ET administration is preferred over intramuscular, if IV and IO routes not available. ET dose is approximately two to three times IV dose.

§ Naloxone administration is not recommended as part of initial resuscitation of neonates in the delivery room. Refer to UpToDate topics on neonatal resuscitation.

¥ AC should be withheld in patients who are sedated and may not be able to protect their airway, unless endotracheal intubation is performed first or naloxone therapy is effective in maintaining an alert state. Endotracheal intubation should not be performed solely for the purpose of giving AC.

Graphic 69375 Version 19.0

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Mark E Ralston, MD, MPH No relevant financial relationship(s) with ineligible companies to disclose. **Susan B Torrey, MD** No relevant financial relationship(s) with ineligible companies to disclose. **James F Wiley, II, MD, MPH** No relevant financial relationship(s) with ineligible companies to disclose.

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